🧊 M003 Week 2 — English Worksheet

Topic: Lines & Rectangles on the Wall (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you use a **loop** to draw a whole **line** of blocks on the wall. Grow **x** to go across, grow **y** to go up. Put four lines together and you get a **rectangle outline** — a picture frame.

1 · Predict 🔮

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
set x to 0
repeat 5 times:
    place stone block at (x, 0)
    add 1 to x
```

Which way does this line grow — across or up? How long is it? Which number stays the same?

```
set y to 0
repeat 4 times:
    place gold block at (3, y)
    add 1 to y
```

This time the line is at x = 3 and y grows. Across or up? How tall is it?

```
set x to 0
repeat 6 times:
    place stone block at (x, 0)
    place stone block at (x, 4)
    add 1 to x
```

Each turn places two blocks at the same x — one at y = 0, one at y = 4. What do the blocks make — one line or two lines? How far apart, top to bottom?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The line should grow **up** at x = 2, from (2, 0) to (2, 4). x should stay 2. Right now the line grows the wrong way.

```
set n to 0
repeat 5 times:
    place stone block at (n, 2)
    add 1 to n
```

Hint: look at which slot n is in — the x slot (across) or the y slot (up).

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The bottom edge of a frame should be **6 blocks long**, from (0, 0) to (5, 0). Right now it comes out one block short.

```
set x to 0
repeat 5 times:
    place gold block at (x, 0)
    add 1 to x
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should make a line of 4 stone blocks growing up from (3, 0). Right now all the blocks land in the **same spot**.

```
set y to 0
repeat 4 times:
    place stone block at (3, y)
```

Hint: read every line of the loop — is anything growing y?

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📸

Now switch to your homework world. Build a **rectangle outline** on the wall — like an empty picture frame — using loops for the four edges (bottom, top, left, right). When you finish, come back here.

Send a photo or video of your frame, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

> First, I …
>
> My bottom edge grew the number … and stayed at y = …
>
> My left edge grew the number … and stayed at x = …
>
> My frame is … wide and … tall.
>
> One tricky moment was when …
>
> If I had more time, I would …

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your frame on the wall. Talk through it like you are teaching someone who has never seen a loop. Try to use these words: **loop**, **line**, **edge**, **across**, **up**.

1. Show your rectangle frame, then show the code that built it.
2. Read one loop out loud and say which number grows and which stays the same.
3. Point at the start and the end of one edge and say their coordinates.
4. Say in your own words why a loop is better than placing every block by hand.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.